

10.18 Consider the data file *mroz* on working wives. Use the 428 observations on married women who participate in the labor force. In this exercise, we examine the effectiveness of a parent's college education as an instrumental variable.

- a. Create two new variables. *MOTHERCOLL* is a dummy variable equaling one if *MOTHEREDUC* > 12, zero otherwise. Similarly, *FATHERCOLL* equals one if *FATHEREDUC* > 12 and zero otherwise. What percentage of parents have some college education in this sample?

父母的大学學歷是否能作為解決受教育程度內生性問題的有效工具變數(IV)的有效性。

(a) 建立新變數並描述統計

新變數定義：

建立虛擬變數 *MOTHERCOLL* (母親受教育年數 > 12 為 1, 否則為 0)
與 *FATHERCOLL* (父親受教育年數 > 12 為 1, 否則為 0)。

統計結果：

- 樣本中母親擁有大學學歷的比例為 12.1495%
- 樣本中父親擁有大學學歷的比例為 11.6822%。

```
> library(POE5Rdata)
> # Select women who participated in the labor force
> mroz_working <- subset(mroz, lfp == 1)
> # Create two new variables: MOTHERCOLL & FATHERCOLL
> mroz_working$mothercoll <- ifelse(mroz_working$mothereduc > 12, 1, 0)
> mroz_working$fathercoll <- ifelse(mroz_working$fathereduc > 12, 1, 0)
> # Calculate percentage
> mean(mroz_working$mothercoll)
[1] 0.1214953
> mean(mroz_working$fathercoll)
[1] 0.1168224
```

- b. Find the correlations between *EDUC*, *MOTHERCOLL*, and *FATHERCOLL*. Are the magnitudes of these correlations important? Can you make a logical argument why *MOTHERCOLL* and *FATHERCOLL* might be better instruments than *MOTHEREDUC* and *FATHEREDUC*?

(b) 相關性分析與邏輯論證

• 相關係數

```
> # Create a correlation matrix of EDUC, MOTHERCOLL, and  
FATHERCOLL
```

```
> cor(mroz_working[c("educ", "mothercoll", "fathercol  
l")]))
```

	educ	mothercoll	fathercoll
educ	1.0000000	0.3594705	0.3984962
mothercoll	0.3594705	1.0000000	0.3545709
fathercoll	0.3984962	0.3545709	1.0000000

• *EDUC* 與 *MOTHERCOLL* 的相關係數為 0.3595。

• *EDUC* 與 *FATHERCOLL* 的相關係數為 0.3985。

• 重要性：中度正相關顯示工具變數與內生變數(教育程度)具備足夠的相關性(Relevance)。

• 邏輯論證：相較於受教育年數，大學學歷虛擬變數可能包含較少的測量誤差，且更能捕捉到「家庭擁有高等教育資源」這一顯著的社經門檻對子女教育的影響。

- c. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* as the instrumental variable. What is the 95% interval estimate for the coefficient of *EDUC*?

(c) 使用單一 IV (*MOTHERCOLL*) 估計薪資方程

- 模型設定：以 *MOTHERCOLL* 作為 *EDUC* 的工具變數進行估計。

```
> # Create a IV regression model (IV: mothercoll)
> mroz1_iv <- ivreg(log(wage)~educ + exper + I(exper^2)|exper +
I(exper^2) + mothercoll, data = mroz_working)
> # See the result
> summary(mroz1_iv)
```

Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
I(exper^2) + mothercoll, data = mroz_working)

Residuals:

	Min	1Q	Median	3Q	Max
	-3.08719	-0.32444	0.04147	0.36634	2.35621

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.1327561	0.4965325	-0.267	0.78932
educ	0.0760180	0.0394077	1.929	0.05440 .
exper	0.0433444	0.0134135	3.231	0.00133 **
I(exper^2)	-0.0008711	0.0004017	-2.169	0.03066 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6703 on 424 degrees of freedom

Multiple R-Squared: 0.147, Adjusted R-squared: 0.1409

Wald test: 8.2 on 3 and 424 DF, p-value: 2.569e-05

```
> # Calculate the 95% CI for educ
> confint(mroz1_iv, "educ", level = 0.95)
```

2.5 % 97.5 %

educ -0.001219763 0.1532557

• 結果：

EDUC 係數的

95%信賴區間為

$[-0.0012, 0.1533]$

• 結論：由於區間包含 0，

代表在 5% 顯著水平

下，教育對薪資的影響

並不顯著。

- d. For the problem in part (c), estimate the first-stage equation. What is the value of the F -test statistic for the hypothesis that $MOTHERCOLL$ has no effect on $EDUC$? Is $MOTHERCOLL$ a strong instrument?

第一階段迴歸檢定(單-IV)

```
> # First stage regression
> first_stage_m <- lm(educ~exper + I(exper^2) + mothercoll,
data = mroz_working)
> # Estimate the first stage equation
> summary(first_stage_m)
```

```
Call:
lm(formula = educ ~ exper + I(exper^2) + mothercoll, data =
mroz_working)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-7.4267 -0.4826 -0.3731  1.0000  4.9353
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.079094   0.303118   39.849 < 2e-16 ***
exper        0.056230   0.042101    1.336  0.182
I(exper^2)   -0.001956   0.001256   -1.557  0.120
mothercoll   2.517068   0.315713    7.973 1.46e-14 ***
```

```
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.133 on 424 degrees of freedom
Multiple R-squared:  0.1347,    Adjusted R-squared:  0.1285
F-statistic: 21.99 on 3 and 424 DF,  p-value: 2.965e-13
```

統計量：檢定 $MOTHERCOLL$
對 $EDUC$ 影响的 F 統計量
為 63.563,

P 值為 1.455×10^{-14} 。

結論： F 值遠大於 10，
拒絕虛無假設，

證明 $MOTHERCOLL$
是一個強工具變數。

```
> # Conduct the hypothesis test
> linearHypothesis(first_stage_m, "mothercoll=0")
```

```
Linear hypothesis test:
mothercoll = 0
```

```
Model 1: restricted model
Model 2: educ ~ exper + I(exper^2) + mothercoll
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	424	1929.9	1	289.32	63.563	1.455e-14 ***

```
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- e. Estimate the wage equation in Example 10.5 using *MOTHERCOLL* and *FATHERCOLL* as the instrumental variables. What is the 95% interval estimate for the coefficient of *EDUC*? Is it narrower or wider than the one in part (c)?

```
> # Create a IV regression model (IV: mothercoll+fathercoll)
> mroz2_iv <- ivreg(log(wage)~educ + exper + I(exper^2)|exper +
I(exper^2) + mothercoll + fathercoll, data = mroz_working)
> # See the result
> summary(mroz2_iv)
```

Call:

```
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
I(exper^2) + mothercoll + fathercoll, data = mroz_working)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-3.07797	-0.32128	0.03418	0.37648	2.36183

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-0.2790819	0.3922213	-0.712	0.47714
educ	0.0878477	0.0307808	2.854	0.00453 **
exper	0.0426761	0.0132950	3.210	0.00143 **
I(exper^2)	-0.0008486	0.0003976	-2.135	0.03337 *

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153, Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06

```
> # Calculate the 95% CI for educ
> confint(mroz2_iv, "educ", level = 0.95)
      2.5 %      97.5 %
educ 0.02751845 0.1481769
```

使用双IV (MOTHERCOLL & FATHERCOLL)
估計

• 結果: 使用兩個IV後, EDUC
係數的95%信賴區間
變為 (0.0275, 0.1482)。

• 比較: 此區間比(c)小題更窄,
顯示加入更多有效的
工具變數提高了估計
的精確度(效率), 且
此時教育的影响變的
顯著。

- f. For the problem in part (e), estimate the first-stage equation. Test the joint significance of *MOTHERCOLL* and *FATHERCOLL*. Do these instruments seem adequately strong?

```
> # First stage regression
> first_stage_mf <- lm(educ~exper + I(exper^2) + mothercoll
+ fathercoll, data = mroz_working)
> # Estimate the first stage equation
> summary(first_stage_mf)
```

```
Call:
lm(formula = educ ~ exper + I(exper^2) + mothercoll + fathercoll,
    data = mroz_working)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-7.2152 -0.3056 -0.2152  0.7627  5.0620
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) 11.890259    0.290251  40.965 < 2e-16 ***
exper        0.049149    0.040133   1.225  0.221
I(exper^2)  -0.001449    0.001199  -1.209  0.227
mothercoll   1.749947    0.322347   5.429 9.58e-08 ***
fathercoll   2.186612    0.329917   6.628 1.04e-10 ***
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 2.033 on 423 degrees of freedom
Multiple R-squared:  0.2161,    Adjusted R-squared:  0.2086
F-statistic: 29.15 on 4 and 423 DF,  p-value: < 2.2e-16
```

```
> # Conduct the hypothesis test
> linearHypothesis(first_stage_mf, c("mothercoll=0", "fathercoll=0"))
```

```
Linear hypothesis test:
mothercoll = 0
fathercoll = 0
```

```
Model 1: restricted model
Model 2: educ ~ exper + I(exper^2) + mothercoll + fathercoll
```

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	425	2219.2				
2	423	1748.3	2	470.88	56.963	< 2.2e-16 ***

```
---
Signif. codes:
  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

一階聯合顯著性檢定 (2IV)

統計量:

聯合檢定的F統計量

為 56.963, p 值 $< 2.2 \times 10^{-16}$

結論: F 值依然大於 10, 顯示這組

工具變數組合具有足夠的強度。

g. For the IV estimation in part (e), test the validity of the surplus instrument. What do you conclude?

```
> # Perform weak instruments, Hausman, and Sargan tests
> summary(mroz2_iv, diagnostics=TRUE)

Call:
ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | exper +
      I(exper^2) + mothercoll + fathercoll, data = mroz_working)

Residuals:
    Min       1Q   Median       3Q      Max
-3.07797 -0.32128  0.03418  0.37648  2.36183

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.2790819   0.3922213   -0.712  0.47714
educ         0.0878477   0.0307808    2.854  0.00453 **
exper        0.0426761   0.0132950    3.210  0.00143 **
I(exper^2)   -0.0008486   0.0003976   -2.135  0.03337 *

Diagnostic tests:
              df1 df2 statistic p-value
Weak instruments  2 423   56.963  <2e-16 ***
Wu-Hausman       1 423    0.519   0.472
Sargan           1  NA    0.238   0.626
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6679 on 424 degrees of freedom
Multiple R-Squared: 0.153,    Adjusted R-squared: 0.147
Wald test: 9.724 on 3 and 424 DF, p-value: 3.224e-06
```

過度識別檢定 (Sargan Test)

• 統計量: Sargan 檢定統計量為 0.238

p 值為 0.626。

• 結論: p 值大於 0.05, 無法拒絕

「工具變數無誤差項不相關」
的虛無假設。

這代表工具變數是有效
且外生的, 模型設定合理。

10.20 The CAPM [see Exercises 10.14 and 2.16] says that the risk premium on security j is related to the risk premium on the market portfolio. That is

$$r_j - r_f = \alpha_j + \beta_j(r_m - r_f)$$

where r_j and r_f are the returns to security j and the risk-free rate, respectively, r_m is the return on the market portfolio, and β_j is the j th security's "beta" value. We measure the market portfolio using the Standard & Poor's value weighted index, and the risk-free rate by the 30-day LIBOR monthly rate of return. As noted in Exercise 10.14, if the market return is measured with error, then we face an errors-in-variables, or measurement error, problem.

- a. Use the observations on Microsoft in the data file *capm5* to estimate the CAPM model using OLS. How would you classify the Microsoft stock over this period? Risky or relatively safe, relative to the market portfolio?

• 資本資產定價模型 (CAPM) 中, 當市場報酬率存在
測量誤差 (Measurement Error) 時, 如何用排名 (RANK)
作為工具更有效地修正偏誤。

```
> # Calculate msft's risk premium
> capm5$y_msft <- capm5$msft - capm5$riskfree
> # Calculate mkt's risk premium
> capm5$x_mkt <- capm5$mkt - capm5$riskfree
> # linear regression model
> msft_ols <- lm(y_msft~x_mkt, data = capm5)
> # See the result
> summary(msft_ols)
```

Call:

```
lm(formula = y_msft ~ x_mkt, data = capm5)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.27424	-0.04744	-0.00820	0.03869	0.35801

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003250	0.006036	0.538	0.591
x_mkt	1.201840	0.122152	9.839	<2e-16 ***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08083 on 178 degrees of freedom

Multiple R-squared: 0.3523, Adjusted R-squared: 0.3486

F-statistic: 96.8 on 1 and 178 DF, p-value: < 2.2e-16

(a) OLS估計與風險分類

• 估計結果: 根據OLS迴歸,
Microsoft的Beta值估計值約
為 1.2018。 > 1

• 風險分類: 由於估計出的 $\beta > 1$,
代表微軟股票的報酬波動
率高於整體市場報酬。

• 結論: 在此期間, 相對於
市場組合, 微軟
股票被歸類為
高風險資產。

- b. It has been suggested that it is possible to construct an IV by ranking the values of the explanatory variable and using the rank as the IV, that is, we sort $(r_m - r_f)$ from smallest to largest, and assign the values $RANK = 1, 2, \dots, 180$. Does this variable potentially satisfy the conditions IV1–IV3? Create $RANK$ and obtain the first-stage regression results. Is the coefficient of $RANK$ very significant? What is the R^2 of the first-stage regression? Can $RANK$ be regarded as a strong IV?

```
> # sort (rm-rf) from smallest to largest
> capm5$rank <- rank(capm5$x_mkt)
> # first stage regression
> first_stage_rank <- lm(x_mkt~rank, data = capm5)
> summary(first_stage_rank)
```

Call:

```
lm(formula = x_mkt ~ rank, data = capm5)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.110497	-0.006308	0.001497	0.009433	0.029513

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.903e-02	2.195e-03	-36.0	<2e-16 ***
rank	9.067e-04	2.104e-05	43.1	<2e-16 ***

Signif. codes:

0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01467 on 178 degrees of freedom

Multiple R-squared: 0.9126, Adjusted R-squared: 0.9121

F-statistic: 1858 on 1 and 178 DF, p-value: < 2.2e-16

(b) 工具變數 RANK 的有效性

• 條件滿足：

IV1 & IV2 (外生性)：使用 RANK

而非原始報酬收益率，可以過濾

掉隨機的衡量誤差，滿足

外生性要求。

• IV3 (相關性)：排名與原始
變數高度相關。

• 統計結果：第一階段迴歸中，RANK 的係數極度顯著 ($P < 2 \times 10^{-16}$)， R^2 高達 0.9126，且 F 統計量為 1858。

• 結論：由於 F 值遠大於 10，

RANK 被視為一個極強的
工具變數。

- c. Compute the first-stage residuals, $\hat{v}$, and add them to the CAPM model. Estimate the resulting augmented equation by OLS and test the significance of $\hat{v}$ at the 1% level of significance. Can we conclude that the market return is exogenous?

```
> library(POESRdata)
> library(AER)
> # Compute the first stage residuals
> capm5$v_hat <- resid(first_stage_rank)
> # Estimate the augmented equation
> augmented_model <- lm(y_msft~x_mkt + v_hat, data = capm5)
> summary(augmented_model)

Call:
lm(formula = y_msft ~ x_mkt + v_hat, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.27140 -0.04213 -0.00911  0.03423  0.34887

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.005984   0.504   0.6146
x_mkt        1.278318   0.126749  10.085 <2e-16 ***
v_hat       -0.874599   0.428626  -2.040  0.0428 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08012 on 177 degrees of freedom
Multiple R-squared:  0.3672,    Adjusted R-squared:  0.36
F-statistic: 51.34 on 2 and 177 DF,  p-value: < 2.2e-16
```

殘差檢定自外生性

- 操作：將一階段殘差加入模型後，其顯著性檢定之 P 值為 0.0428。

- 結論：在 1% 顯著水準下 ($0.0428 > 0.01$)，無法拒絕虛無假設。因此，在 1% 水準下可認定市場報酬率為外生。

- d. Use *RANK* as an IV and estimate the CAPM model by IV/2SLS. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

```
> #-----
> #part d
>
> # 2SLS model: use rank as an IV
> rank_iv <- ivreg(y_msft~x_mkt | rank, data = capm5)
> summary(rank_iv)

Call:
ivreg(formula = y_msft ~ x_mkt | rank, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.271625 -0.049675 -0.009693  0.037683  0.355579

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003018   0.006044   0.499   0.618
x_mkt        1.278318   0.128011   9.986 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08092 on 178 degrees of freedom
Multiple R-Squared:  0.3508,    Adjusted R-squared:  0.3472
Wald test: 99.72 on 1 and 178 DF,  p-value: < 2.2e-16

> #-----
```

IV/2SLS 估計與比較

- IV 估計值：使用單一 IV (*RANK*) 得到的 Beta 值為 1.2783。
 - 比較與預期：此值高於 OLS 的 1.2018。
- 這符合預期，因為衡量誤差會導致「稀釋偏誤」，使 OLS 估計值往零收縮；IV 方法成功修正了此偏誤，還原出較大的 Beta 值。

- e. Create a new variable $POS = 1$ if the market return ($r_m - r_f$) is positive, and zero otherwise. Obtain the first-stage regression results using both $RANK$ and POS as instrumental variables. Test the joint significance of the IV. Can we conclude that we have adequately strong IV? What is the R^2 of the first-stage regression?

```
> # Create a new variable: POS
> capm5$pos <- ifelse(capm5$x_mkt > 0, 1, 0)
>
> # first stage regression with rank and pos
> first_stage_rank_pos <- lm(x_mkt~rank + pos, data = capm5)
> summary(first_stage_rank_pos)
```

```
Call:
lm(formula = x_mkt ~ rank + pos, data = capm5)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.109182 -0.006732  0.002858  0.008936  0.026652
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0804216  0.0022622   -35.55  <2e-16 ***
rank         0.0009819  0.0000400    24.55  <2e-16 ***
pos         -0.0092762  0.0042156    -2.20   0.0291 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.01451 on 177 degrees of freedom
Multiple R-squared:  0.9149,    Adjusted R-squared:  0.9139
F-statistic: 951.3 on 2 and 177 DF,  p-value: < 2.2e-16
```

双工具变量 (RANK & POS) 检验

• 统计结果

第一階段聯合顯著檢定的F
統計量為951.3，其 $R^2 = 0.9149$ 。

- 結論：F值極高，代表 RANK 与 POS
組合是非常強的工具變數組合。

- f. Carry out the Hausman test for endogeneity using the residuals from the first-stage equation in (e). Can we conclude that the market return is exogenous at the 1% level of significance?

```
> #part f
>
> # Compute the first stage residuals
> capm5$v_hat_2 <- resid(first_stage_rank_pos)
>
> # Estimate the augmented equation
> augmented_model_2 <- lm(y_msft~x_mkt + v_hat_2, data = capm5)
> summary(augmented_model_2)
```

```
Call:
lm(formula = y_msft ~ x_mkt + v_hat_2, data = capm5)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.27132 -0.04261 -0.00812  0.03343  0.34867
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.003004   0.005972   0.503   0.6157
x_mkt        1.283118   0.126344  10.156 <2e-16 ***
v_hat_2      -0.954918   0.433062  -2.205  0.0287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.07996 on 177 degrees of freedom
Multiple R-squared:  0.3696,    Adjusted R-squared:  0.3625
F-statistic: 51.88 on 2 and 177 DF,  p-value: < 2.2e-16
```

Hausman 内生性檢定

• 結果：使用(e)殘差進行的檢定P值為0.0287。

• 結論：在1%顯著水準下
(0.0287 > 0.01)，依然
無法拒絕外生性假設，
維持市場報酬率為外生
的結論。

- g. Obtain the IV/2SLS estimates of the CAPM model using *RANK* and *POS* as instrumental variables. Compare this IV estimate to the OLS estimate in part (a). Does the IV estimate agree with your expectations?

```
> #-----
> #part g
>
> # 2SLS model: use rank and pos as IVs
> rank_pos_iv <- ivreg(y_msft~x_mkt | rank + pos, data = capm5)
> summary(rank_pos_iv)
```

```
Call:
ivreg(formula = y_msft ~ x_mkt | rank + pos, data = capm5)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.27168	-0.04960	-0.00983	0.03762	0.35543

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.003004	0.006044	0.497	0.62
x_mkt	1.283118	0.127866	10.035	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08093 on 178 degrees of freedom

Multiple R-Squared: 0.3507, Adjusted R-squared: 0.347

Wald test: 100.7 on 1 and 178 DF, p-value: < 2.2e-16

```
> #-----
```

(g) 双IV之估计结果

IV估计值：同时使用RANK
与POS得到的Beta值为1.2831。

結論：此估计值与(d)小题

相似且大于OLS，再次

证实IV法有效修正

了测量误差带来的

低估问题。

- h. Obtain the IV/2SLS residuals from part (g) and use them (not an automatic command) to carry out a Sargan test for the validity of the surplus IV at the 5% level of significance.

```
> #part h
>
> # Obtain the IV/2SLS residuals
> e_hat_iv <- resid(rank_pos_iv)
>
> # Sargan test: see whether rank and pos can predict error
> aux_model <- lm(e_hat_iv ~ rank + pos, data = capm5)
> summary(aux_model)

Call:
lm(formula = e_hat_iv ~ rank + pos, data = capm5)

Residuals:
    Min       1Q   Median       3Q      Max
-0.26914 -0.04702 -0.00801  0.03771  0.35674

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.0022220  0.0126326  -0.176   0.861
rank         0.0001370  0.0002234   0.613   0.540
pos         -0.0174499  0.0235409  -0.741   0.460

Residual standard error: 0.08103 on 177 degrees of freedom
Multiple R-squared:  0.003103, Adjusted R-squared:  -0.008162
F-statistic: 0.2754 on 2 and 177 DF, p-value: 0.7596

>
> # R^2
> r2 <- summary(aux_model)$r.squared
>
> # Calculate NxR^2
> N <- nrow(capm5)
> sargan_statistic <- N*r2
> print(sargan_statistic)
[1] 0.5584634
>
> # Chi-Square test
> #df = L-B = 1
> # Calculate the critical value
> critical_value <- qchisq(0.95, df = 1)
>
> if (sargan_statistic > critical_value){
+   cat("Reject H0: at least one of the IVs is correlated with error")
+ } else{
+   cat("Fail to reject H0: The two IVs are valid!")
+ }
Fail to reject H0: The two IVs are valid!
> |
```

Sargan 过度识别检定

- 統計量：手动計算出的 Sargan 統計量為 0.5585。

- 結論：由於統計量遠小於卡方分佈臨界值 (3.841)，無法拒絕虛無假設，結論是剩餘的工具變數是有效的，且工具變數組合與誤差項不相關。

当解釋變數(市場報酬率)存在雜訊時，OLS 會低估資產的風險 (Beta 值)。透過建立 RANK 這類非原始變數高度相關但誤差無關的工具變數，可以更精準地捕捉到微觀股票真實的市場風險。